

Hasse Diagrams



A Hasse diagram is a graphical rendering of a partially ordered set displayed via the cover relation of the POSET with an implied upward orientation. A point is drawn for each element of the poset, and line segments are drawn between these points according to the following two rules:

1. If $x < y$ in the poset, then the point corresponding to x appears lower in the drawing than the point corresponding to y .
2. The line segment between the points corresponding to any two elements x and y of the poset is included in the drawing iff x covers y or y covers x .

Hasse diagrams are also called upward drawings.

Hasse Diagrams



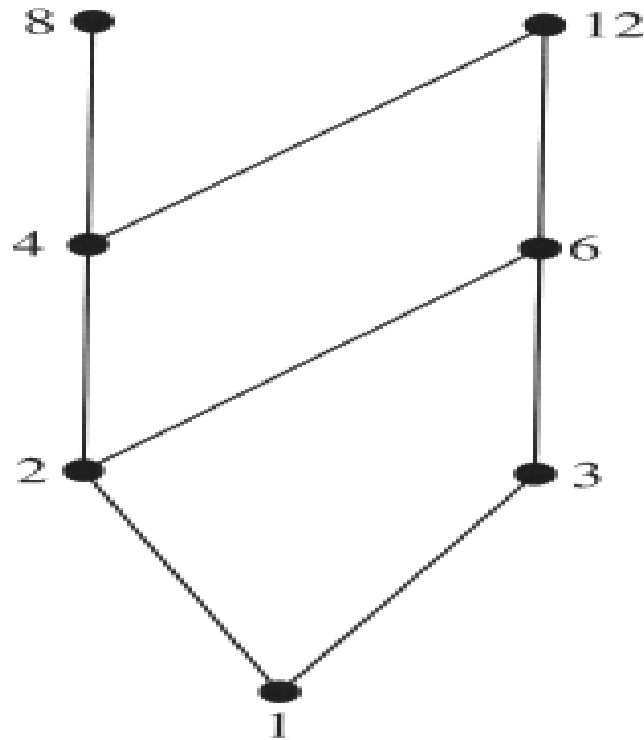
Given any partial order relation defined on a finite set, it is possible to draw the directed graph so that all of these properties are satisfied.

Remove self loop and transitive edge.

This makes it possible to associate a somewhat simpler graph, called a *Hasse diagram*, with a partial order relation defined on a finite set.

Hasse Diagram

- For the poset $(\{1,2,3,4,6,8,12\}, |)$



Construct the Hasse diagram of $(P(\{a, b, c\}), \subseteq)$.

The elements of $P(\{a, b, c\})$ are

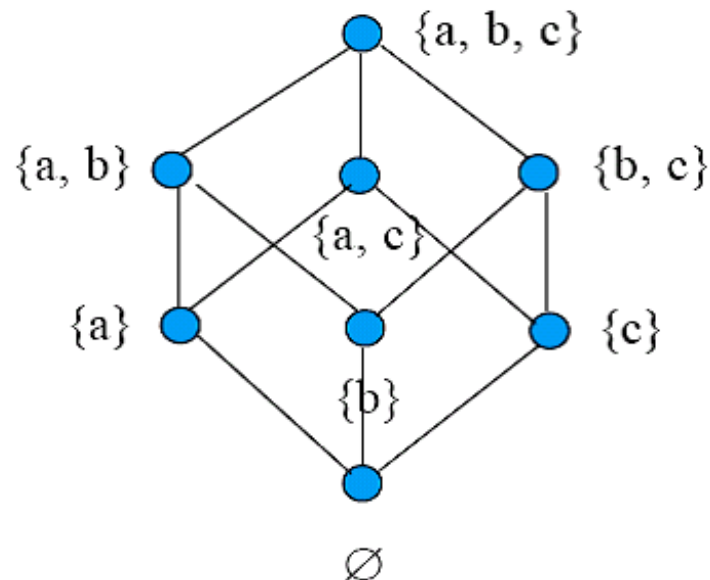
$\emptyset$

$\{a\}, \{b\}, \{c\}$

$\{a, b\}, \{a, c\}, \{b, c\}$

$\{a, b, c\}$

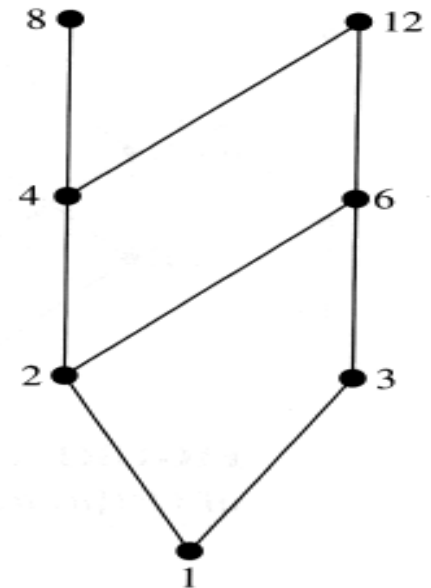
The digraph is



Maximal and Minimal Elements

a is a *maximal* in the poset $(S, \preceq)$ if there is no $b \in S$ such that $a \prec b$. Similarly, an element of a poset is called *minimal* if it is not greater than any element of the poset. That is, a is *minimal* if there is no element $b \in S$ such that $b \prec a$.

It is possible to have multiple minimals and maximals.



Consider a poset $(A, \leq)$

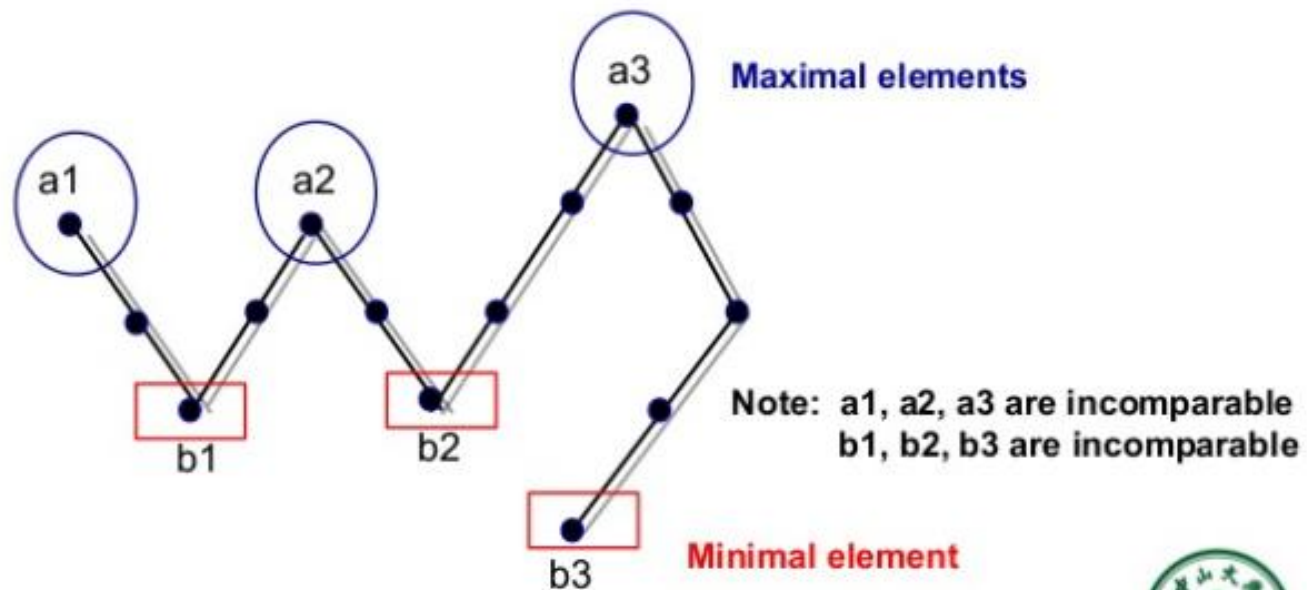
- Maximal Element

An element a in A is called a maximal element of A if there is no element c in A such that $a \leq c$.

- Minimal Element

An element b in A is called a minimal element of A if there is no element c in A such that $c \leq b$.

- Example: Find the maximal and minimal elements in the following Hasse diagram



- Example

Let A be the poset of nonnegative real number with the usual partial order $\leq$. Then 0 is a minimal element of A . There are no maximal elements of A

- Example

The poset Z with the usual partial order $\leq$ has no maximal elements and has no minimal elements

■ Theorem 1

If $(A, \leq)$ be a poset, then

1. If greatest element exists, then it is unique.
2. If least element exists, then it is unique.

Proof:

Assume that there are two greatest elements of poset $(A, \leq)$, say a and b .

Therefore, $x \leq a$ and $x \leq b$, $\forall x \in A$.

$\therefore a \leq b$ (b is greatest element)

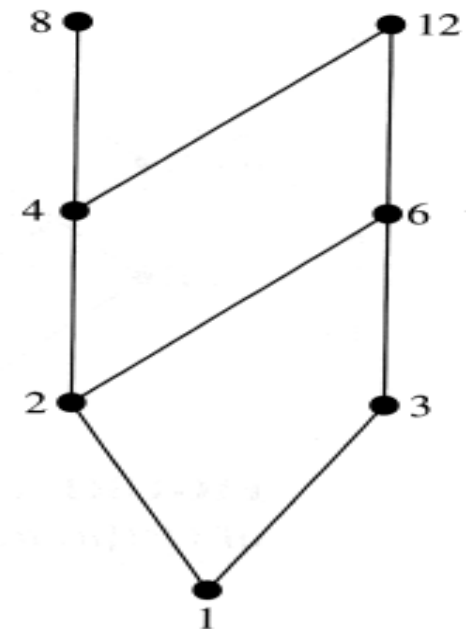
and $b \leq a$ (a is greatest element)

$\therefore$ by antisymmetric property, $a = b$.

Similarly, for least element.

Greatest Element and Least Element

a is the *greatest element* in the poset $(S, \preceq)$ if $b \preceq a$ for all $b \in S$. Similarly, an element of a poset is called the *least element* if it is less or equal than all other elements in the poset. That is, a is the *least element* if $a \preceq b$ for all $b \in S$.



Upper bound, Lower bound



Sometimes it is possible to find an element that is greater than all the elements in a subset A of a poset $(S, \preceq)$. If u is an element of S such that $a \preceq u$ for all elements $a \in A$, then u is called an **upper bound** of A . Likewise, there may be an element less than all the elements in A . If l is an element of S such that $l \preceq a$ for all elements $a \in A$, then l is called a **lower bound** of A .

Upper Bound of:

$$ub\{2,6\}=\{6,12\}$$

$$ub\{4,8\}=\{8\}$$

$$ub\{2,6\}=\{6,12\}$$

$$Ub\{3,4\}=\{12\}$$

$$Ub\{8,12\}=\text{no upper bound}$$

$$Ub\{2,3,6\}=\{6,12\}$$

$$Ub\{1,2,4\}=\{4,8,12\}$$

Lower Bounds:

$$lb\{2,6\}=\{1,2\}$$

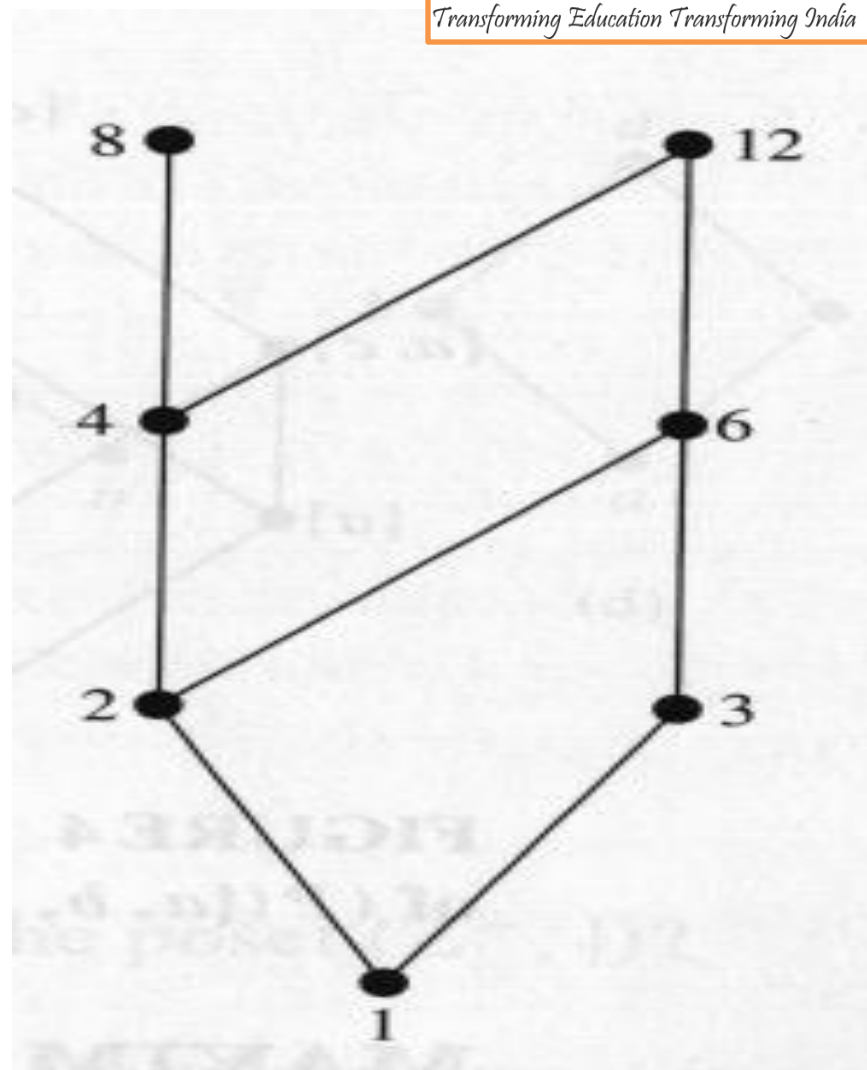
$$lb\{4,8\}=\{1,2,4\}$$

$$lb\{3,4\}=\{1\}$$

$$lb\{8,12\}=\{4,2,1\}$$

$$lb\{2,3,6\}=\{1\}$$

$$lb\{1,2,4\}=\{1,2\}$$



Least Upper Bound, Greatest Lower Bound



The element x is called the *least upper bound* (lub) of the subset A if x is an upper bound that is less than every other upper bound of A .

The element y is called the *greatest lower bound* (glb) of A if y is a lower bound of A and $z \preceq y$ whenever z is a lower bound of A .

- In the poset $(P(S), \subseteq)$, $\text{lub}(A, B) = A \cup B$. What is the $\text{glb}(A, B)$?

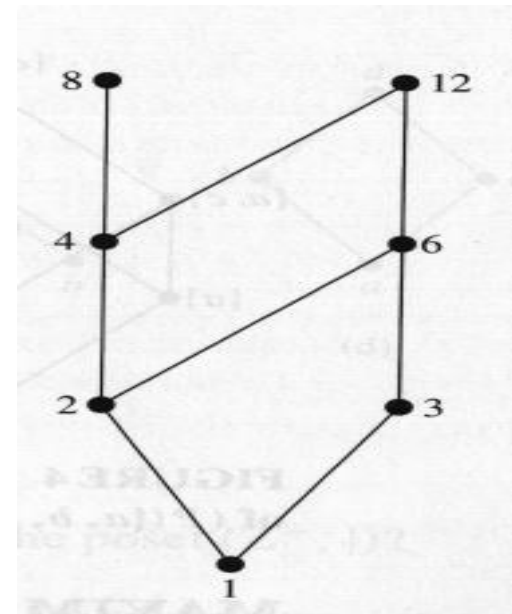
Upper Bound and least upper bound of:

$\text{ub}\{2,6\}=\{6,12\}$ $\text{lub}\{2,6\}=\{6\}$
 $\text{ub}\{4,8\}=\{8\}$ $\text{lub}\{4,8\}=\{8\}$
 $\text{ub}\{2,6\}=\{6,12\}$
 $\text{Ub}\{3,4\}=\{12\}$
 $\text{Ub}\{8,12\}=\text{no upper bound}$
 $\text{Ub}\{2,3,6\}=\{6,12\}$ $\text{lub}\{2,3,6\}=\{6\}$
 $\text{Ub}\{1,2,4\}=\{4,8,12\}$ $\text{lub}\{1,2,4\}=\{4\}$

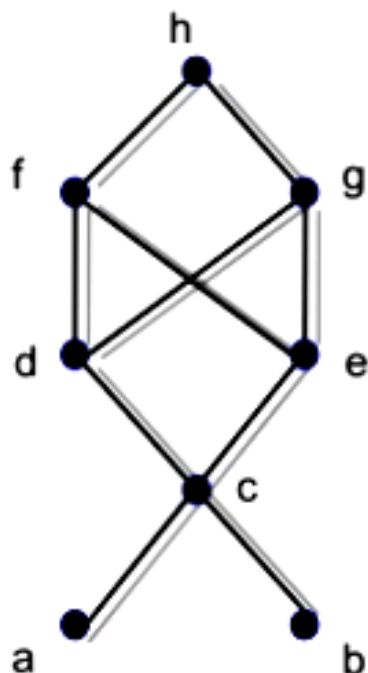
Lower Bounds and greatest lower bounds:

$\text{lb}\{2,6\}=\{1,2\}$ $\text{glb}\{2,6\}=\{2\}$
 $\text{lb}\{4,8\}=\{1,2,4\}$ $\text{glb}\{4,8\}=\{4\}$

$\text{lb}\{3,4\}=\{1\}$	$\text{glb}\{3,4\}=\{1\}$
$\text{lb}\{8,12\}=\{4,2,1\}$	$\text{glb}\{8,12\}=\{4\}$
$\text{lb}\{2,3,6\}=\{1\}$	$\text{glb}\{2,3,6\}=\{1\}$
$\text{lb}\{1,2,4\}=\{1,2\}$	



Find all least upper bounds and all greatest lower bounds of (a) $B_1 = \{a, b\}$ (b) $B_2 = \{c, d, e\}$



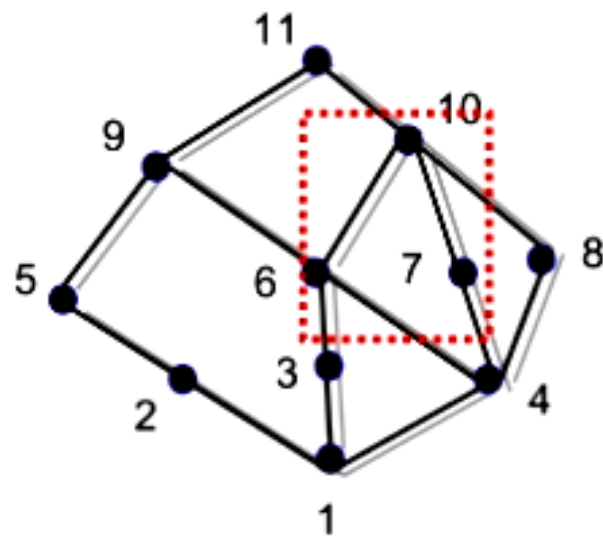
(a) Since B_1 has no lower bounds, it has no greatest lower bounds; However,
 $LUB(B_1) = c$

(b) Since the lower bounds of B_2 are c, a and b, we find that $GLB(B_2) = c$.
The upper bounds of B_2 are f, g and h. Since f and g are not comparable, we conclude that B_2 has no least upper bound.



■ Example

Let $A = \{1, 2, 3, \dots, 11\}$ be the poset whose Hasse diagram is shown below. Find the LUB and GLB of $B = \{6, 7, 10\}$, if they exist.



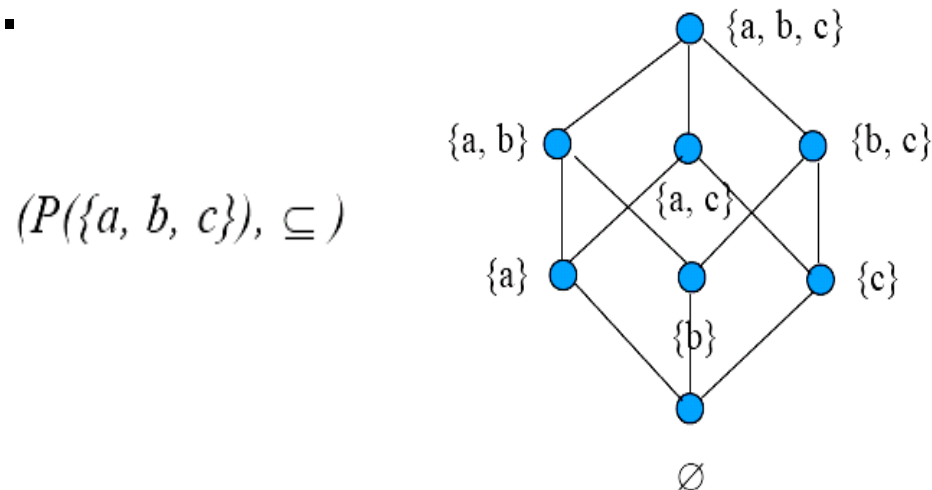
The upper bounds of B are 10, 11, and
 LUB(B) is 10 (the first vertex that can be
 Reached from $\{6, 7, 10\}$ by upward paths)

8 The lower bounds of B are 1, 4, and
 GLB(B) is 4 (the first vertex that can be
 Reached from $\{6, 7, 10\}$ by downward
 paths)



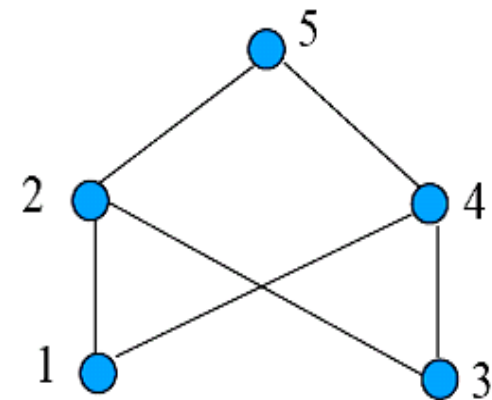
Lattices

A partially ordered set in which *every pair* of elements has both a least upper bound and a greatest lower bound is called a *lattice*.



Consider the elements 1 and 3.

- Upper bounds of 1 are 1, 2, 4 and 5.
- Upper bounds of 3 are 3, 2, 4 and 5.
- 2, 4 and 5 are upper bounds for the pair 1 and 3.
- There is no lub since
 - 2 is not related to 4
 - 4 is not related to 2
 - 2 and 4 are both related to 5.
- There is no glb either.



The poset is not a lattice.

Examples 21 and 22, p. 575 in Rosen.